Module 4 – Exercise 2: Looking Up a Word in the Dictionary

Steven Francis · SDEV120 · Flowchart built in Lucidchart

The task

Draw a structured flowchart of the process of looking up a word in the dictionary, using the values at the top and end of each page. You open the dictionary somewhere, check the guide words, and flip forward or back until you reach the page that should hold the word.

How the flowchart works

- The user enters targetWord. lowPage and highPage start as the first and last pages of the dictionary; onPage starts as "N".
- Search loop: while onPage = "N" AND lowPage <= highPage, open or flip to a page between lowPage and highPage and read its guide words – topWord (top of the page) and bottomWord (end of the page).
- If targetWord comes before topWord, we went too far: highPage = currentPage – 1 (flip back).
- Otherwise, if targetWord comes after bottomWord, we have not gone far enough: lowPage = currentPage + 1 (flip forward).
- Otherwise the word belongs on this page, so onPage = "Y" and the loop ends.
- The page range shrinks on every pass, so if lowPage passes highPage the word cannot be in the dictionary, and the program says so.
- If the right page was found, a second loop reads entries until it reaches targetWord or runs out of entries. If entryWord = targetWord, the word and its definition are displayed; if not, the program reports that the word is not in the dictionary.

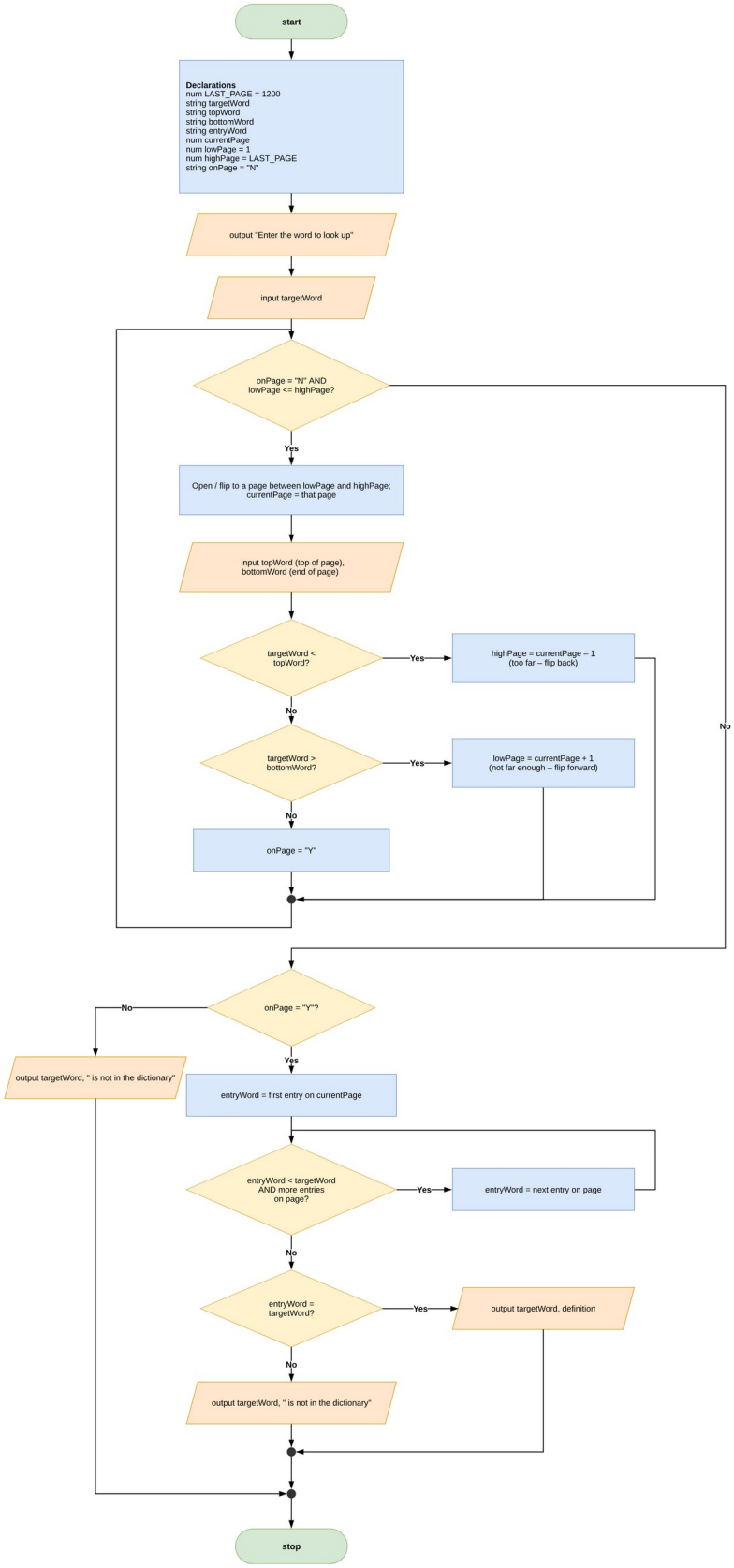
Structures used

The flowchart is structured: it uses only the three basic structures, and each structure has one entry and one exit.

- **Sequence** – steps run one after another, top to bottom.
- **Selection** – a diamond tests a Boolean expression and takes exactly one Yes/No path; both paths rejoin at a single point (the small black dot).
- **Loop** – the test sits at the top of the loop; while it is true the body repeats, then control returns to the same test.

Symbol key: green rounded = start/stop, blue rectangle = process, orange parallelogram = input/output, yellow diamond = decision.

Flowchart



Structures used
- **Sequence** – steps run one after another, top to bottom.
- **Selection** – each diamond tests a Boolean expression and takes exactly one Yes/No path; both paths rejoin at a single point.
- **Loop** – a test at the top of the loop; while it is true the body repeats, then control returns to the same test.

Symbol key
Green rounded = start / stop
Blue rectangle = process
Orange parallelogram = input / output
Yellow diamond = decision

Guide words
topWord = first word printed at the top of the page; bottomWord = last word at the end of the page. If targetWord comes before topWord we went too far (flip back); if it comes after bottomWord we have not gone far enough (flip forward). lowPage/highPage shrink the range each time, so if they cross the word is not in the dictionary.